

ON THE NORMALIZING CONSTANT FOR THE BINGHAM-VON MISES-FISHER MATRIX DISTRIBUTION

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SUMMARY

Let $X(p \times m)$, $m \leq p$, be a random matrix defined on the Stiefel manifold $X'X = I_m$, then the density $C \exp(\text{tr}(A(X-m)B(X-M)'))$ of X generalizes the Bingham distribution and the von Mises-Fisher distribution. This paper deals with the normalizing constant of the density and various special cases of the density. Special attention has been given to the case $m=1$. The first two moments of X and the estimation of the parameters has been considered in this case.

1. Introduction

Let $X(p \times m)$, $m \leq p$, be a random matrix defined on the Stiefel manifold $X'X = I_m$, denoted by $O(p, m)$. Let dX denote the Haar measure on $O(p, m)$, then the area of the Stiefel manifold is

$$V_{p,m} = \int_{O(p,m)} dX = 2^m \pi^{\frac{1}{2}pm} / \Gamma_m(\frac{1}{2}p).$$

The normalized measure will be denoted by

$$(1.1) \quad (dX) = dX / V_{p,m}$$

The von Mises-Fisher matrix distribution of X considered by Khatri and Mardia (1977) is defined as

$$(1.2) \quad (F_1(Z))^{-1} \exp(\text{tr}ZX), X \in O(p, m)$$

where the normalizing constant is

$$\begin{aligned}
 (1.3) \quad F_1(Z) &= \int_{O(p,m)} \exp(\text{tr}(ZX)) (dX) \\
 &= {}_0F_1\left(\frac{1}{2}p; \frac{1}{2}ZZ', Z(m \times p)\right).
 \end{aligned}$$

As mentioned by the authors, a generalization of the Bingham distribution (Bingham (1974)) will be

$$(1.4) \quad (F_2(A))^{-1} \exp(\text{tr}(AXX')), X \in O(p,m)$$

where

$$\begin{aligned}
 (1.5) \quad F_2(A) &= \int_{O(p,m)} \exp(\text{tr}(AXX')) (dX) \\
 &= {}_1F_1\left(\frac{1}{2}m; \frac{1}{2}p; A\right), A(p \times p) \text{ nonsingular and symmetric} \\
 &\quad (\text{n.s.s.})
 \end{aligned}$$

An extension to (1.2) and (1.4) is mentioned briefly in Khatri and Mardia (1975, 1977) and will be considered in more detail here. The normalizing constant given by these authors differ from the one given in this paper. The representations differ since their constant is based on an extension of the method of Kotz, Johnson and Boyd (1967) while the constant considered here is based on an extension by Crowther (1975) of the polynomial given by Hayakawa (1972). Special attention will be given to the spherical case in the last section. The case treated by Bingham and Mardia (1978) also follows from this case.

2. Preliminary results

From Crowther (1975) the following lemma can be formulated:

Lemma 2.1

If $X(p \times m)$, $m \leq p$, is distributed multivariate normal $N(M; \Sigma_1 \otimes \Sigma_2)$ with mean $M(p \times m)$ and covariance $\Sigma_1 \otimes \Sigma_2$, $\Sigma_1 (p \times p)$ and $\Sigma_2 (m \times m)$, then the density of $S = X'X$ is given by

$$\begin{aligned}
 (2.1) \quad & (2^{\frac{1}{2}mp} \Gamma_m(\frac{1}{2}p))^{-1} |\Sigma_1|^{-\frac{1}{2}m} |\Sigma_2|^{-\frac{1}{2}p} \exp(-\frac{1}{2} \text{tr}(\Sigma_1^{-1} M \Sigma_2^{-1} M')) \\
 & \exp(-\frac{1}{2} q \text{tr}(\Sigma_2^{-1} S)) |S|^{(p-m-1)/2} \sum_{k=0}^{\infty} \Sigma_K((\frac{1}{2}p)_K k!)^{-1} \\
 & P_K(-\frac{1}{\sqrt{2}} \Sigma_2^{-\frac{1}{2}} M' \Sigma_1^{-\frac{1}{2}} (I_p - q \Sigma_1)^{-\frac{1}{2}} \Sigma_1^{-1} - q I_p, \frac{1}{2} \Sigma_2^{-\frac{1}{2}} S \Sigma_2^{-\frac{1}{2}}).
 \end{aligned}$$

$q \geq 0$ is a fixed constant such that $I_p - q \Sigma_1$ is positive definite. P_K is a polynomial

defined by

$$(2.2) \quad P_K(T, A, B) = \pi^{-\frac{1}{2}mp} \int_U \exp(-\text{tr}(U + iT)(U + iT)') C_K(-BUAU') dU,$$

for $T(m \times p)$ complex, $U(m \times p)$ real and $A(p \times p)$ and $B(m \times m)$ real symmetric. C_K being a zonal polynomial for the partition K of k . Explicit expressions for $P_K(T, A, B)$, $k \leq 3$, were given by the author.

Remark

It is interesting to note from (2.2) that if $Y(p \times m)$, $m \leq p$, is distributed $N(O; I_p \otimes I_m)$, then for $A(p \times p)$ symmetric

$$(2.3) \quad E C_K(AYY') = 2^k (\frac{1}{2}m)_K C_K(A).$$

3. Generalized Bingham-von Mises-Fisher matrix distribution

Let $S = I_m$, $\Sigma_1 = A^{-1}$ and $\Sigma_2 = -\frac{1}{2}B^{-1}$ in (2.1) and using (1.1), it follows that

$$\begin{aligned} F(A, B, M) &= \int_{O(p, m)} \exp(\text{tr}(A(X-M)B(X-M)')) (dX) \\ &= \exp(\text{tr}(AMB M')) \exp(\text{qtr}(B)) \sum_{k=0}^{\infty} \Sigma_K \\ &\quad ((\frac{1}{2}p)_K k!)^{-1} P_K(iB^{\frac{1}{2}} M' A^{\frac{1}{2}} (I - qA^{-1})^{-\frac{1}{2}}, A - qI_p, -B) \end{aligned}$$

We therefore can state the following theorem:

Theorem 1

If $X(p \times m)$, $m \leq p$, is distributed as the Generalized Bingham-von Mises-Fisher matrix distribution with parameters A, B and M , then the density of X is given by

$$(3.1) \quad (F(A, B, M))^{-1} \exp(\text{tr} A(X-M)B(X-M)'), X \in O(p, m)$$

where

$$\begin{aligned} (3.2) \quad F(A, B, M) &= \exp(\text{tr} AMB M') \exp(\text{qtr} B) \sum_{k=0}^{\infty} \Sigma_K \\ &\quad ((\frac{1}{2}p)_K k!)^{-1} P_K(iB^{\frac{1}{2}} M' A^{\frac{1}{2}} (I - qA^{-1})^{-\frac{1}{2}}, A - qI_p, -B). \end{aligned}$$

The density (3.1) can also be considered as a conditional density on the Stiefel manifold from a multivariate normal distribution.

The following densities of X follow as special cases from (3.1):

(i) $M = O$: Generalized Bingham matrix distribution.

$$(3.3) \quad (F(A, B, O))^{-1} \exp(\text{tr}(AXBX')), X \in O(p, m)$$

where

$$(3.4) \quad \begin{aligned} F(A, B, O) &= \int_{O(p, m)} \exp(\text{tr}(\text{tr}(AXBX'))) (dX) \\ &= \exp(q \text{tr}(B)) \sum_{k=0}^{\infty} \Sigma_K(\frac{1}{2}p)_K k!^{-1} P_K(O, A - qI, B). \end{aligned}$$

After further simplification, it follows that

$$F(A, B, O) = \exp(q \text{tr}(B)) {}_0F_0(A - qI, B)$$

(ii) $M = O, B = I_m$: Bringham matrix distribution.

$$(3.5) \quad (F(A, I_m, O))^{-1} \exp(\text{tr}(AXX')), X \in O(p, m)$$

where

$$(3.6) \quad F(A, I_m, O) = \exp(mq) {}_1F_1(\frac{1}{2}m; \frac{1}{2}p; A - qI)$$

By taking $q = O$, (1.5) follows

(iii) $A = I_p, B = I_m$: von Mises-Fisher matrix distribution.

This distribution has been studied by Khatri and Mardia (1977) and the density is

$$(3.7) \quad (F(I_p, I_m, M))^{-1} \exp(\text{tr}(M'M) \exp(m) \exp(\text{tr}(-2M'X))), X \in O(p, m).$$

By direct integration

$$(3.8) \quad \begin{aligned} F(I_p, I_m, M) &= \int_{O(p, m)} \exp(-2 \text{tr}(M'X)) (dX) \\ &= {}_0F_1(\frac{1}{2}p; M'M). \end{aligned}$$

This result also follows from (3.2) by assuming $q=O$. (3.8) is similar to the normalizing

constant given by Khatri and Mardia (1977).

$$(3.9) \quad F(I_p, I_m, M) = \exp(\text{tr}(M'M)) \sum_{k=0}^{\infty} \sum_K ((\frac{1}{2}p)_K k!)^{-1} P_K(iM', I_p, -I_m) \\ = \exp(\text{tr}(MM')) \sum_{k=0}^{\infty} \sum_K ((\frac{1}{2}p)_K k!)^{-1} L_K^{\frac{1}{2}(p-m-a)}(-M'M)$$

L_K is the Laguerre polynomial defined by Constantine (1966) as

$$L_K^{\frac{1}{2}(p-m-1)}(-M'M) = (\frac{1}{2}p)_K C_K(I_m) \sum_{j=0}^K \sum_J \binom{K}{j} C_J(M'M) ((\frac{1}{2}p)_J C_J(I_m))^{-1}.$$

For the relation between P_K and L_K , see Khatri (1977). Hence from (3.8) and (3.9),

$$(3.10) \quad \sum_{k=0}^{\infty} \sum_K ((\frac{1}{2}p)_K k!)^{-1} L_K^{\frac{1}{2}(p-m-1)}(-M'M) = \exp(m) {}_0F_1(\frac{1}{2}p; M'M).$$

As a check on (3.10), let $M = 0$, using the fact $L_K^{\frac{1}{2}(p-m-1)}(0) = (\frac{1}{2}p)_K C_K(I_m)$.

(iv) $A = I_p$: von Mises-Fisher matrix distribution.

$$(3.11) \quad (F(I_p, B, M))^{-1} \exp(\text{tr}(B)) \exp(\text{tr}(MBM')) \exp(-2\text{tr}(MBX')), X \in O(p, m)$$

where

$$(3.12) \quad F(I_p, B, M) = \exp(\text{tr}(qB)) \exp(\text{tr}(MBM')) \sum_{k=0}^{\infty} \sum_K ((\frac{1}{2}p)_K k!)^{-1} \\ P_K(iB^{\frac{1}{2}}M' (I-q)^{-\frac{1}{2}}, (1-q)I_p, -B).$$

Alternatively from the integral (3.8)

$$(3.13) \quad F(I_p, B, M) = \exp(\text{tr}(B)) \exp(\text{tr}(MBM')) {}_0F_1(\frac{1}{2}p; M'B^2M).$$

Comparing (3.12) and (3.13) the identity

$$(3.14) \quad \sum_{k=0}^{\infty} \sum_K ((\frac{1}{2}p)_K k!)^{-1} P_K(iB^{\frac{1}{2}}M' (I-q)^{-\frac{1}{2}}, (1-q)I_p, -B) \\ = \exp((1-q)\text{tr}(B)) {}_0F_1(\frac{1}{2}p; M'B^2M) \text{ follows}$$

(v) $B = I_m$: Bingham-von Mises-Fisher Matrix distribution.

$$(3.15) \quad (F(A, I_m, M))^{-1} \exp(\text{tr} A(X-M)(X-M')), X \in O(p, m)$$

where

$$(3.16) \quad F(A, I_m, M) = \exp(\text{tr}(AMM')) \exp(mq) \sum_{k=0}^{\infty} \sum_K ((\frac{1}{2}p)_K k!)^{-1} P_K(iM'A^{\frac{1}{2}} \\ (I - qA^{-1})^{-\frac{1}{2}}, A - qI_p, -I_m).$$

A very interesting case is when $m=1$ and $B=b$ a fixed constant. This case will be called the Bingham-von Mises-Fisher distribution and will be treated in the next section.

4. The Bingham-von Mises-Fisher distribution

Since the Bingham-von Mises-Fisher distribution can also be considered as a conditional distribution on a unit sphere from a multivariate normal, we shall let $m=1$, $b=-\frac{1}{2}$ and $A=\Sigma^{-1}$ in (3.1). As a corollary from theorem 1 it therefore follows that;

Corollary 1

If $X(p \times 1)$ is distributed as a multivariate normal with mean μ and covariance Σ , then the conditional density of $X \in S_p$ is given by

$$(4.1) \quad (F(\Sigma^{-1}, -\frac{1}{2}, \mu))^{-1} \exp(-\frac{1}{2}(x-\mu)' \Sigma^{-1} (x-\mu)), x \in S_p$$

where

$$(4.2) \quad F(\Sigma^{-1}, -\frac{1}{2}, \mu) = \exp(-\frac{1}{2}\mu' \Sigma^{-1} \mu) \sum_{k=0}^{\infty} ((\frac{1}{2}p)_k k!)^{-1} P_k(-\frac{1}{\sqrt{2}} \mu' \Sigma^{-\frac{1}{2}}, \Sigma^{-1}, \frac{1}{2}).$$

The polynomial P_k is given by (see Crowther (1975)).

$$(4.3) \quad P_k(-\frac{1}{\sqrt{2}} \mu' \Sigma^{-\frac{1}{2}}, \Sigma^{-1}, \frac{1}{2}) \\ = (-\frac{1}{2})^k E \{ (v + \frac{1}{\sqrt{2}} \Sigma^{-\frac{1}{2}} \mu)' \Sigma^{-1} (v + \frac{1}{\sqrt{2}} \Sigma^{-\frac{1}{2}} \mu) \}^k \\ = (-\frac{1}{2})^k \mu_k(\mu, \Sigma), \quad \text{say}$$

where $v(p \times 1)$ has the density $\pi^{-\frac{1}{2}p} \exp(-v'v)$, $v \in R^p$

According to theorem 1, the distribution (4.1) will also be called the Bingham-von Mises-Fisher distribution with parameters Σ^{-1} , $-\frac{1}{2}$ and μ .

The k -th moment about the origin $\mu_k(\mu, \Sigma)$ can be expressed in terms of the cumulants $\kappa_j(\mu, \Sigma)$, $j=1, \dots, k$ (see Kendall and Stuart (1958)). For instance

$$\mu_0(\mu, \Sigma) = 1, \mu_1(\mu, \Sigma) = \kappa_1(\mu, \Sigma), \mu_2(\mu, \Sigma) = \kappa_2(\mu, \Sigma) + \kappa_1^2(\mu, \Sigma).$$

The j -th cumulant $\kappa_j(\mu, \Sigma)$ is given by Crowther (1975) as

$$(4.4) \quad \kappa_j(\mu, \Sigma) = \frac{1}{j} (j-1)! \operatorname{tr} \Sigma^{-j} \mu \mu' \Sigma^{-(j+1)} \mu.$$

The calculation of the normalizing constant

$$(4.5) \quad F(\Sigma^{-1}, -\frac{1}{2}, \mu) = \exp(-\frac{1}{2} \mu' \Sigma^{-1} \mu) \sum_{k=0}^{\infty} ((-\frac{1}{2})^k / (\frac{1}{2} p)_k k!) \mu_k(\mu, \Sigma)$$

has been programmed and can be calculated quite accurately.

4.1 The characteristic function of moments of $X(px1)$

The characteristic function of $X(px1)$ distributed according to (4.1) is given by

$$\begin{aligned} (4.6) \quad \phi_X(t) &= E \exp(it'X) \\ &= (F(\Sigma^{-1}, -\frac{1}{2}, \mu))^{-1} \int_{S_p} \exp(it'x - \frac{1}{2}(x-\mu)' \Sigma^{-1}(x-\mu)) (dx) \\ &= (F(\Sigma^{-1}, -\frac{1}{2}, \xi) / F(\Sigma^{-1}, -\frac{1}{2}, \mu)) \exp(it' \mu - \frac{1}{2} t' \Sigma t) \end{aligned}$$

where $\xi = \mu + i \Sigma t$, t a $px1$ vector and $(dx) = dx/V_{p,1}$.

The first two moments of X can now be derived, namely

$$E(X) = i^{-1} \partial \phi_X(t) / \partial t|_{t=0}$$

where

$$\begin{aligned} (4.7) \quad \partial \phi_X(t) / \partial t &= \exp(it' \mu - \frac{1}{2} t' \Sigma t) / F(\Sigma^{-1}, -\frac{1}{2}, \mu) \{ i F(\Sigma^{-1}, -\frac{1}{2}, \xi) \xi + \partial F(\Sigma^{-1}, -\frac{1}{2}, \xi) / \partial t \} \end{aligned}$$

$$\begin{aligned} (4.8) \quad \partial F(\Sigma^{-1}, -\frac{1}{2}, \xi) / \partial t &= i F(\Sigma^{-1}, -\frac{1}{2}, \xi) \xi + \exp(-\frac{1}{2} \xi' \Sigma^{-1} \xi) \sum_{k=0}^{\infty} ((-\frac{1}{2})^k / (\frac{1}{2} p)_k k!) \sum_{j=1}^k (\partial \mu_k(\xi, \Sigma) / \partial \kappa_j) \end{aligned}$$

$$(\xi, \Sigma)) (\partial \kappa_j(\xi, \Sigma) / \partial t).$$

From Kendall and Stuart (1958)

$$\partial \mu_k(\xi, \Sigma) / \partial \kappa_j(\xi, \Sigma) = \binom{k}{j} \mu_{k-j}(\xi, \Sigma)$$

and by direct differentiation

$$\partial \kappa_j(\xi, \Sigma) / \partial t = -ij! \Sigma^{-j} \xi.$$

Let

$$(4.9) \quad \nu(\xi, \Sigma) = \sum_{k=0}^{\infty} ((-1)^k / (\frac{1}{2}p)_k k!) \mu_k(\xi, \Sigma)$$

and

$$(4.10) \quad \nu^{(1)}(\xi, \Sigma) = \partial \nu(\xi, \Sigma) / \partial \xi = - \sum_{k=0}^{\infty} ((-1)^k / (\frac{1}{2}p)_k k!) \sum_{j=1}^k \binom{k}{j} j! \mu_{k-j}(\xi, \Sigma) \Sigma^{-(j+1)} \xi.$$

Then from (4.7), (4.8) and (4.9)

$$(4.11) \quad \begin{aligned} \partial \phi_X(t) / \partial t \\ = i \Sigma \exp(it' \mu - \frac{1}{2} t' \Sigma^{-1} t) \nu^{(1)}(\xi, \Sigma) / \nu(\mu, \Sigma). \end{aligned}$$

Hence

$$(4.12) \quad \nu = E(X) = \Sigma \nu^{(1)}(\mu, \Sigma) / \nu(\mu, \Sigma).$$

As a measure of dispersion $R^2 \approx \nu' \nu$ can be used.

4.2 Estimates of μ and Σ .

Let $X_1, \dots, X_n$ denote a random sample of size n from the Bingham-von Mises-Fisher distribution (4.1). The log likelihood function is given by

$$(4.13) \quad \ln L(\mu, \Sigma) = -n \ln F(\Sigma^{-1}, -\frac{1}{2}, \mu) - \frac{1}{2} \sum_{i=1}^n (X_i - \mu)' \Sigma^{-1} (X_i - \mu).$$

Maximizing with respect to μ , we obtain

$$(4.14) \quad \Sigma \nu^{(1)}(\mu, \Sigma) / \nu(\mu, \Sigma) = \bar{X}.$$

$$\text{where} \quad \bar{X} = \frac{1}{n} \sum_{j=1}^n X_j.$$

This implies that

$$(4.15) \quad \hat{\nu} = \bar{X}$$

is a ML estimator for ν given in (4.12).

Maximizing (4.13) with respect to Σ^{-1} the following equation is obtained:

$$(4.16) \quad \nu^{(2)}(\mu, \Sigma) / \nu(\mu, \Sigma) + \frac{1}{2} S = \frac{1}{2} \mu \bar{X}' + \frac{1}{2} \bar{X} \mu'$$

$$\text{where} \quad S = \frac{1}{n} \sum_{j=1}^n X_j X_j',$$

$$\begin{aligned} \nu^{(2)}(\mu, \Sigma) &= \partial \nu(\mu, \Sigma) / \partial \Sigma^{-1} \\ &= \sum_{k=0}^{\infty} ((-\frac{1}{2})^k / (\frac{1}{2} p)_k k!) \sum_{j=1}^k \binom{k}{j} j! \mu_{k-j}(\mu, \Sigma) G_j \end{aligned}$$

and

$$G_j = \frac{1}{2} \Sigma^{-(j-1)} - \frac{1}{2} \Sigma_{m=1}^{j+1} \Sigma^{(1-m)} \mu \mu' \Sigma^{(m-1-j)}.$$

Solving for μ and Σ iteratively from (4.14) and (4.16) estimates can be obtained if a random sample is given. This problem as well as tests of hypotheses will be dealt with in a later communication. Interesting to note is that $\bar{X}$ and S are sufficient statistics for estimating μ and Σ or $\Sigma^{-1}(\Sigma^{-1} + \xi I_p)^{-1} \mu$ and $(\Sigma^{-1} + \xi I_p)^{-1}$ respectively for arbitrary ξ since the parameters are not unique.

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